

Pizza Shop Billing System Using Python

Abstract

This mini project is a simple **Pizza Shop Billing System** developed using **Python programming language**.

The main objective of this project is to display the pizza menu, allow the user to select pizzas, enter quantity, and calculate the total bill amount.

The program uses **functions** and **dictionary data structure** to store menu details and perform operations efficiently.

This project helps beginners understand basic Python concepts like functions, dictionaries, conditional statements, and user input handling.

Objectives

- To design a simple pizza ordering system
 - To use **dictionary** for storing pizza details
 - To implement **functions** for modular programming
 - To calculate the total bill based on user order
 - To improve logical thinking and Python basics
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System Requirements

Hardware Requirements

- Computer / Laptop
- Minimum 2GB RAM

Software Requirements

- Python 3.x
 - Any IDE (IDLE / PyCharm / VS Code)
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Tools & Technologies Used

- **Programming Language:** Python
- **Data Structure:** Dictionary

- **Concepts Used:**
 - Functions
 - Conditional Statements
 - User Input
 - Loops (optional extension)
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Project Description

The Pizza Shop Billing System allows the user to:

1. View the available pizza menu
2. Select a pizza from the menu
3. Enter the quantity
4. Calculate and display the total bill amount

The pizza details like name and price are stored using a **dictionary**, and different operations are handled using separate **functions**.

Methodology

1. Create a dictionary to store pizza name and price
 2. Create a function to display the menu
 3. Create a function to take order details from the user
 4. Calculate total bill using price × quantity
 5. Display order details and total amount
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Algorithm

1. Start
2. Display menu options
3. Ask user to choose menu or order pizza
4. If menu selected → show menu
5. If order selected →

- Ask pizza number
 - Ask quantity
 - Calculate total bill
6. Display total amount
 7. Stop

Flow Chart (Text Representation)

Start



Display Menu Options



User Choice



Show Menu / Order Pizza



Enter Quantity



Calculate Bill



Display Total Amount



End

Source Code

```
menu = {  
    1: ("Veg Pizza", 200),  
    2: ("Cheese Pizza", 250),  
    3: ("Paneer Pizza", 300)
```

```
}
```

```
def show_menu():
```

```
    print("---- MENU ----")
```

```
    for key, value in menu.items():
```

```
        print(key, ".", value[0], "-", value[1], "Rs")
```

```
def order_pizza():
```

```
    show_menu()
```

```
    choice = int(input("Enter pizza number (1-3): "))
```

```
    qty = int(input("Enter quantity: "))
```

```
    if choice in menu:
```

```
        name, price = menu[choice]
```

```
        total = price * qty
```

```
        print("Pizza Name:", name)
```

```
        print("Quantity:", qty)
```

```
        print("Total Amount:", total)
```

```
    else:
```

```
        print("Invalid choice")
```

```
def pizza_shop():
```

```
    print("1. Show Menu")
```

```
    print("2. Order Pizza")
```

```
    choice = int(input("Enter your choice: "))
```

```
if choice == 1:
    show_menu()
elif choice == 2:
    order_pizza()
else:
    print("Invalid option")
```

```
pizza_shop()
```

Sample Output

1. Show Menu

2. Order Pizza

Enter your choice: 2

---- MENU ----

1. Veg Pizza - 200 Rs

2. Cheese Pizza - 250 Rs

3. Paneer Pizza - 300 Rs

Enter pizza number: 1

Enter quantity: 2

Pizza Name: Veg Pizza

Quantity: 2

Total Amount: 400

Advantages

- Easy to understand and use
 - Reduces manual billing errors
 - Beginner-friendly Python project
 - Modular code using functions
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Limitations

- Only one order at a time
 - No database connection
 - No GST or discount calculation
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Future Enhancements

- Add multiple pizza orders
 - Include GST and discounts
 - Store order details in a file/database
 - Add graphical user interface (GUI)
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Conclusion

This mini project successfully demonstrates a **Pizza Shop Billing System** using Python. It effectively uses **functions** and **dictionary** to manage menu and billing operations. The project is suitable for beginners to understand core Python concepts and can be extended further with advanced features.